



Converting Spatial to Social: Using Persistent Homology to Understand Social Groups

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Abstract

In social settings, people display sophisticated spatial behaviors—for example, one might naturally enter into a conversation by sidling up to a group. Artificial agents will need the ability to reason about spatial representations of social information to understand not only how social groups form, but also how to interact within and around them. Leveraging the insight that people reason about shared space topologically rather than geometrically, we employ techniques from applied topology to introduce a new method for social group analysis that improves quantifiability and enables rigorous analysis of social group structure. We present a novel topological mathematical formalism called the *social simplicial complex* that provides an equivalence relation for socially analogous configurations of people and is provably robust against small perturbations and noise. Moreover, this formalism suggests quantifiable metrics to assess the confidence of social group existence and the social closeness of people within groups. We further use this formalism to introduce an open-source toolkit for evaluating possible models of social relationships, which we name the Social Topological Analysis (SoTA) Toolkit. Finally, we explore algebraic topology’s potential to serve more generally as a powerful tool for multi-modal social data processing, and its possibilities for further applications in social-spatial analysis.

CCS Concepts

• **Human-centered computing** → HCI theory, concepts and models; • **Mathematics of computing** → Algebraic topology; • **Computing methodologies** → Spatial and physical reasoning;

Keywords

Social Groups, F-Formations, Topology, Persistent Homology, Robotics

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1 Introduction

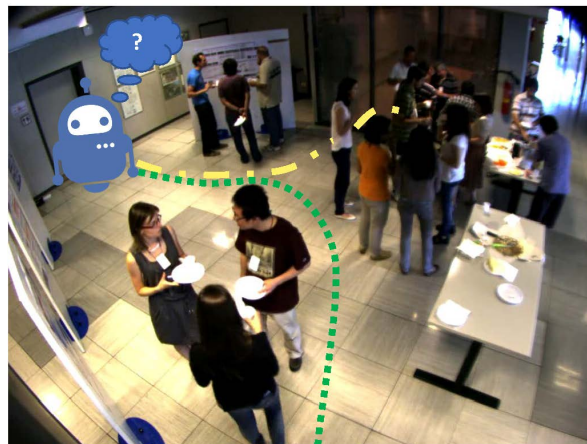


Figure 1: A scene from the SALSA dataset [1] showing possible trajectories for a robot to either navigate through participants (green) or join a group (yellow).

Humans, as inherently social beings, are masters of gathering and harmonizing a profusion of information when interacting with others in social settings. For example, one may read minute shifts in body orientation to determine where best to sidle up to a conversational group, perform exaggerated turns when walking with a group to maintain focus on the speaking member, or assess the density of people in a room to determine how close to stand next to a stranger without having to strike up conversation. But how would a roboticist, for example, design a server robot that fluidly navigates around conversational groups and charismatically offers hors d’oeuvres to those who seem interested? To enable algorithms to align with human expectations in social settings, we require a means to computationally convert the disparate multi-modal inputs

humans consider, such as speech or gesture, into quantitative measures of social relationships. Access to such quantitative measures empowers developers to mathematically analyze social scenes, generate simulated social data, and design algorithms for embodied agents to interact with humans in real-world social environments.

However, computational understanding of social scenes is challenging. Social and spatial information are deeply linked; people are often able to signal evolutions in social group formations without words [22] as simple body language shifts can suffice [16]. Thus, the analysis of social groups requires both a spatial lens and the ability to disentangle social from spatial.

Social-spatial understanding itself relies on latent social signals implicit across a vast amount of raw data. Task-specific constraints (e.g., avoiding vs. approaching individuals) may affect the relevance of different latent signals, and unpredictable factors such as density of individuals can introduce spurious information. Additionally, it is not feasible to gather data representative of every potential salient scenario, and real-world social data is limited to the hardware available at the moment of collection. This results in inconsistencies across datasets and deployment platforms. To address these challenges, we require an abstraction that can reduce high-dimensional, multi-modal, and noisy inputs to a useful lower-dimensional representation containing only salient information.

Our key insight is that *topology is naturally suited for abstracting social-spatial information*. Humans intuitively reason about spaces topologically [21, 34] and similarly use topological language to communicate spatial information to each other (e.g., giving the direction ‘go left past the door’ instead of providing specific coordinates) [25]. Furthermore, in many social settings, requirements for preserving social grouping are best reasoned about topologically—for example, a group can be passed on either side [25, 26].

In this paper, we contribute:

- (1) A mathematical structure called a *social simplicial complex* that permits quantifiable measures of interpersonal relationships and social groups by leveraging techniques from topological data analysis’s persistent homology. Furthermore, we show that the structure is deterministic, provides guarantees for resilience to noise, and equates topological invariance to social invariance (Section 3).
- (2) The *Social Topological Analysis (SoTA) Toolkit*, an open-source toolkit that incorporates the social simplicial complex to enable analysis and comparison of social relationship models (e.g., modeling the exclusivity of shared space between individuals). We also provide a concrete use case of persistent homology on real-world data and show how this toolkit enables quick, flexible iteration for designing functions and tuning parameters. Our code is available at <https://github.com/vk-chen/SoTA-Toolkit>.
- (3) A discussion on the potential of topological data analysis and algebraic topology for social-spatial interaction research.

2 Related Work

Non-verbal cues are integral to human communication. For example, people can use eye gaze [31, 37, 48], gesture [23], and positioning of their head, torso, and feet to communicate anything from willingness to interact [31, 48] to recognition of aggression [9]. Thus, it is

crucial to consider various modalities of input in order to generate the most natural social interaction.

There are several challenges to computationally understanding human social interaction. First, there is a vast number of inputs—gestures, head pose, body pose, etc.—and their combinations to consider [51]. Second, noise from imperfect collection of real-world data compounds in the multi-modal setting, drastically decreasing model accuracy when large amounts of noise are present [27]. Third, data is expensive to collect, and ground truth labeling for large datasets is not tenable if it requires manual labeling [20]. As a result, systems depend highly on specific data inputs, making it difficult to transfer insights from one setting or dataset to another.

One example domain that would benefit from transfer across setting-specific models is the study of social-spatial groups, a well-established area of interest for communities such as multi-modal interaction, human robot interaction, and computer vision. Existing work has tackled problems including group detection [19, 35, 42, 49, 50], understanding spatial group configurations [2, 10, 14, 47], and social navigation [30]. However, the social spaces across these works vary, and these prior methods are tailored to their specific social settings. For example, social group scenarios can be partitioned into “static” scenes and “dynamic” scenes [43], such as cocktail party conversational groups versus pedestrian groups in a mall. Literature suggests that people’s spatial configurations vary across these two settings [22, 33]. A robot deployed in an environment where there are both walking and stationary social groups would require a general model capable of handling both. There may also be insights generally applicable to human social behavior that can be shared across both domains—for example, people tend to make eye contact with the speaker of a group regardless of whether they are walking or not. The ability to explicitly compare and contrast disparate models of social relationships would enable researchers to unify insights from prior work to better understand social-spatial group interactions. In this paper, we provide a topological abstraction that provides the abilities to unify and compare.

Why Topology? Topology serves as a means to reduce spatial geometric information into a lower-dimensional space that aligns with human intuition. Prior work has shown that people default to reasoning about space topologically rather than geometrically, especially when reasoning about social information [21, 34]. Consider signs in an airport [25]: directions are shown through pictograms or arrows, which are easier and more useful for people to navigate by compared to geometrically precise maps. Enabling machines to reason in topological terms increases their potential for socially-fluent operation in human spaces.

We demonstrate the power of persistent homology to address some of the challenges listed above—e.g., handling the multi-modal nature of social information (Section 3) and being resilient to noise (Section 3.2)—and present our open-source SoTA Toolkit (Section 4). We evaluate the SoTA Toolkit by comparing foundational theories from the static social group and dynamic social group communities, demonstrating the power of persistent homology to unify insights from prior literature (Section 5). Finally, we discuss how our methods can be used for generation of labeled data (Section 6).

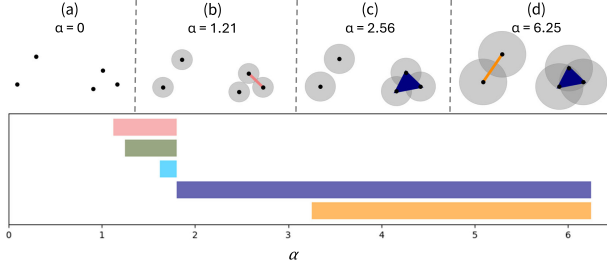


Figure 2: Top: An example (a–d) of increasing the radius of α -balls to determine connectivity between points in 2D ($\alpha \sim r^2$ [28]). An edge (1-simplex) is formed in (b), and a face (2-simplex) is formed in (c) that subsumes the previous edges of the triangle. Bottom: A persistence barcode diagram. Colors correspond to the edge or face (top) each barcode is computed for. The green and light blue bars do not appear in the top animation as they both appear and disappear (when the purple triangle is formed) between (b) and (c). Configuration (c) is most likely for this toy data because the purple triangle has the longest barcode—the highest persistence value.

3 The Mathematical Structure of Social Groups

Interpreting social information topologically lends rigor to computational modeling of social information, provides provable guarantees, and introduces quantitative measures that enables both the ranking of groups in a scene and comparison across models. This mathematical lens yields abstractions that are compact, flexible, and useful across applications.

We introduce a mathematical formalism that allows us to equate “topological invariance” to “social invariance.” In other words, if two scenes produce topologically equivalent structures under this formalism, they are also socially equivalent regardless of geometric variation. This connection between topology and sociality unlocks the ability to computationally and quantifiably understand the social semantics of a scene.

3.1 Background: Persistent Homology

In applied mathematics, *topological data analysis (TDA)* is used to capture underlying topological structures from data [4]. A primary component of TDA is *persistent homology*.

Persistent homology “tak[es]... individual points and assembl[es] them into a coherent image of a continuum—it infers the continuous from the discrete” [53], capturing the general shape of spatial data. This ability aligns with the human ability to infer something continuous (e.g., paths that avoid interrupting a conversation) from something discrete (e.g., people positioned in space). Persistent homology reveals the underlying “shape” and structure of spatial data. Consider a dataset of phone tower ping locations [8]. In the same way one may “squint their eyes” [32] to see the high-level shape of this point cloud, persistence allows us to quantifiably identify the areas of coverage and “holes” in the overall network.

3.1.1 Alpha Complexes. A simplicial complex is a set made of points, line segments, triangles, and n -dimensional triangle equivalents. An alpha complex is a type of simplicial complex that is guaranteed to capture the underlying topology of a dataset [7]. In

this work, we use alpha complex construction to create a deterministic topological structure for social group relationships.

Consider a set of points in space, as shown in Figure 2. To capture connectivity between points, we grow a ball around each point. An $(n-1)$ -simplex is formed when n points’ balls connect. For example, two intersecting balls create an edge (1-simplex) in (b), and three create a triangle (2-simplex) in (c) [7], as illustrated on the top of Figure 2. On the bottom of Figure 2, we include a corresponding “persistence barcode diagram,” a diagram used in TDA to visualize the formation and subsumption of edges and faces as α grows. For example, the pink edge is the first to be connected, forming between (a) and (b), and is subsumed by the purple triangle in (c). Thus, the pink bar in the barcode diagram appears the earliest of all the bars and disappears when the purple bar begins.

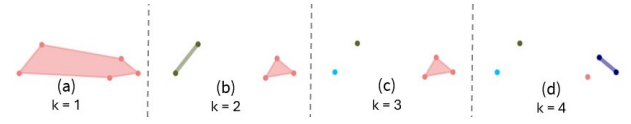


Figure 3: Applying k -means clustering to the toy data from Figure 2 for $k = \{1, 2, 3, 4\}$ (a - d). (b) $k = 2$ matches Figure 2(d), (c) $k = 3$ matches Figure 2(c), and (d) $k = 4$ matches Figure 2(b). Note that k -means clustering provides no measure of group likelihood.

Intuitively, the most basic version of persistent homology used in this paper can be compared to k -means clustering. k -means clustering groups data into k clusters based on relative proximity (Figure 3). Persistence, however, sorts by how close subsets of points are and orders by when connected components emerge. To capture the same information that persistence does using k -means instead, one would have to re-run clustering for all k from 1 to the number of points in the dataset. Even then, the results would be non-deterministic with no ordering to the proposed groups.

3.1.2 Alpha Complexes for Social Information. Alpha complexes are able to handle unstructured data, are deterministic, and are robust to small amounts of noise. Given that multimodal data can be unstructured, and real-world social-spatial data is noisy from people naturally shifting around and measurements made by imperfect sensors, these advantages make alpha complexes particularly appropriate for analyzing social information.

To construct our mathematical formalism, called the “social simplicial complex,” we require as input an existing “social relationship function” that defines possible ways people may be socially linked (such as one that outputs a shared social space of pairs given poses). We refer to the original data about agents in the scene as $\mathbf{P}$, the “social relationship function” as $f(\mathbf{P})$, and the higher-order function we provide that creates the social simplicial complex as $S(f(\mathbf{P}))$.

The social relationship function $f(\mathbf{P})$ transforms the data $\mathbf{P}$ from the (possibly multimodal) input space to a vector representation that can then be used as input to S . The function requires that the input must include spatial information about people in a scene (e.g., positions, raw images, etc.), and that the output representation contains relevant social information (e.g., relative poses of pairs of people or how closely people in a triad are spaced). f must be

injective, but the vector representation can be of any finite dimensionality. So long as $f(\mathbf{P})$ satisfies these requirements, then the function S is well-defined and captures salient social information. In this section, we show the higher-order function $S(f(\mathbf{P}))$ that maps $f(\mathbf{P})$ to its:

- **Social simplicial complex:** a mathematical structure that captures social spatial information $S(f(\mathbf{P}))$;
- **Social cohesion:** the quantifiable measure for understanding which subsets of people are more socially engaged with each other; and
- **Social persistence:** the quantifiable measure of the confidence in a groups' true existence, or the robustness of groups' underlying topology to geometric noise.

Social Simplicial Complex: Let $\mathbf{P}$ be a set of people. A “social simplicial complex” $S(f(\mathbf{P}))$ is an alpha complex generated from the output of a function $f(\mathbf{P})$ that converts raw multi-modal data into social-spatial relationships. If two configurations' complexes have the same underlying topology, they lie in the same equivalence class [4] and thus contain the same social information w.r.t. $f(\mathbf{P})$.

3.1.3 Filtrations. From this social simplicial complex, we can determine which sets of people are the most socially connected by observing the process of α -balls with radius r growing as described in Figure 2, where α is computed using r according to [28]. Despite the balls growing continuously, new edges and faces are formed only at discrete steps. We index these discrete α values as a_i for the i -th index. For each a_i , we obtain a different simplicial complex K_i . Organizing the simplicial complexes K_i according to their a_i value, we obtain a “filtration” F : a sequence of simplicial complexes with each a subset of the following complex. $F : \emptyset = K_0 \subseteq K_1 \subseteq \dots, K_i \subseteq \dots \subseteq K_n = K$ [4]. For example, the set of simplicial complexes shown growing from (a) to (d) in Figure 2 is a valid filtration. A function of the filtration's α -ball size at a_i (such as used in [28] is referred to as a filtration value.

Social Cohesion: A set of people is more “socially cohesive” than another if their corresponding sub-simplicial complex has a lower filtration value. The smaller the “social distance”—the distance in the underlying vector space—between people, the more likely they are to be in a group.

This “social cohesion” value can be used to examine a scene such as in Figure 4, where more than one possible interpretation of the configuration exists. Subgroups with lower (corresponding to better) cohesion values are more close-knit. For example, if the

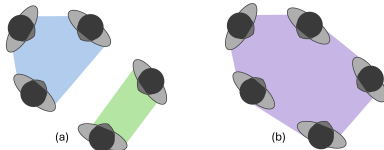


Figure 4: An example of five people in a scene that could ambiguously comprise either a three person group plus a two-pair (a) or a five person group (b).

configuration in Figure 4 (a) better matches the ground truth than that of Figure 4 (b), the simplicial complexes corresponding to the two indicated subgroups in (a) should have better “social cohesion” than those corresponding to other arrangements of group members. **Birth, Death, and Persistence:** The “persistence” of an n -simplex, or the difference in filtration value from when the simplex appears to when it disappears as α increases, provides crucial information about the likelihood of the social sub-simplicial complex corresponding to a true group rather than being an artifact of noise. The greater the distance between the “birth” and “death” of a simplex, the more persistent it is. In other words, it exists uninterrupted for a longer duration when growing α and is thus more likely to be a true underlying topological structure [6]. For example, in Figure 2, the line segment between the two rightmost points exists only for timestep (b) before it is subsumed by the triangle (face) connecting the three rightmost points, which exists for both timesteps (c) and (d). This group corresponds to the longest bar (purple) in its persistence barcode diagram Figure 2. Therefore, the triangle is more “persistent” and a better candidate for the true underlying group.

Social Persistence: Social groups are more “socially persistent” if their corresponding sub-simplicial complexes are present for a longer stretch of α growth.

3.2 Equating Topological Equivalence to Social Equivalence

Here, we show that if $f(\mathbf{P})$ accurately reflects social closeness, then $S(f(\mathbf{P}))$ is guaranteed to capture that as well.

THEOREM 1. *If it is true that distance in $f(\mathbf{P})$ is an accurate metric for “social closeness,” then in $S(f(\mathbf{P}))$, lower (better) “social cohesion” scores capture more tightly-knit groups.*

PROOF. Let $\mathbf{P}$ be a set of people's configurations and $f(\mathbf{P})$ be a function for “social closeness.” Let $U = f(\mathbf{P})$ with $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d} \in U$.

Assume $|\mathbf{a} - \mathbf{b}| < |\mathbf{c} - \mathbf{d}|$ and the value of $S_{a,b}$ (the α where the edge a, b is born) is higher than the cohesion value of $S_{c,d}$, $\alpha_{a,b}$ and $\alpha_{c,d}$ respectively. If $\alpha_{a,b} > \alpha_{c,d}$ then $2\alpha_{a,b} > 2\alpha_{c,d}$. $2\alpha_{a,b} = |\mathbf{a} - \mathbf{b}|$ and $2\alpha_{c,d} = |\mathbf{c} - \mathbf{d}|$ by definition. This implies $|\mathbf{a} - \mathbf{b}| > |\mathbf{c} - \mathbf{d}|$, which is a contradiction to the assumption that $|\mathbf{a} - \mathbf{b}| < |\mathbf{c} - \mathbf{d}|$. $\square$

3.3 Resilience to Noise

Social persistence inherits persistent homology's resilience to noise [5]. The stability theorem guarantees that “small changes in the function imply only small changes in the [persistence] diagram” [3]. As a result, the topological features of the social simplicial complex are robust to small amounts of geometric noise, which we demonstrate with real data in Section 5.3. The full stability theorem for simplicial filtrations can be found in [5].

4 Application of Social Simplicial Complex: Open-Source Social Topological Analysis (SoTA) Toolkit

The use of persistent homology for social group analysis is both theoretically grounded and practically useful for researchers working with multi-modal representations of social information. It can

handle broad social concepts ranging from gauging social dislike to determining the speaker of a group. Here, we contribute the Social Topological Analysis (SoTA) Toolkit, an open-source toolkit leveraging the social simplicial complex that system developers can use to examine, design, and optimize their models.

Consider, for example, a roboticist tasked with identifying social groups to navigate around [18, 36, 52], or deciding best point of entry for a robot to join a conversation [35]. They may construct a model, test on an existing dataset, then work towards deployment. Deploying a real robot is difficult and costly, so it is impractical to iterate on social relationship models using hardware platforms. It is possible to test in simulation, but performance is upper-bounded by the accuracy of the simulation to real-life behavior.

The social simplicial complex structure (Section 3) allows us to interpret social equivalence across topologically equivalent simplicial complexes (Section 3.2). Since social cohesion and social persistence are provably correct and deterministic, the social simplicial complex enables troubleshooting of proposed models. Since $S(f(\mathbf{P}))$ is a well-defined higher-order function and can additionally be mapped back to its original input values in $f(\mathbf{P})$, we can additionally use $S(f(\mathbf{P}))$ to re-examine properties of $f(\mathbf{P})$.

Consider two sets of people $\mathbf{P}$ and $\mathbf{P}'$ with ground truth labels of social groups available (e.g., Alameda-Pineda et al. [1]). If the resulting $S(f(\mathbf{P}))$ and $S(f(\mathbf{P}'))$ do not lie in the same equivalence class, the discrepancy suggests that $f(\mathbf{P})$ does not accurately model social closeness.

This result allows roboticists to use a social simplicial complex as a means for designing the best function $f(\mathbf{P})$ for their scenario. We open-source this code in the SoTA Toolkit, available at <https://github.com/vk-chen/SoTA-Toolkit>.

4.1 SoTA Toolkit for Function Selection

Assume the function designer has two models f and g that both are meant to capture social dynamics of people in a scene. These models are not easily comparable in an apples to apples fashion—for example, one model uses eye gaze while the other does not.

Given a scene of people $\mathbf{P}$, the designer can use our SoTA Toolkit and generate social simplicial structures using $\mathbf{P}$ for both f and g .

The SoTA Toolkit outputs simplicial complexes that are subsets of the overall social simplicial complex along with corresponding social cohesion and social persistence scores. With these scores, the designer can sort candidate groupings by how socially cohesive or how socially persistent they are.

A perfect function should return a list of social groups that precisely reflect $\mathbf{P}$'s ground truth. We can compare f and g 's proposed group rankings against the ground truth to determine the superior model. We walk through a concrete example in Section 5.

After selecting a model, the designer can also use the SoTA Toolkit to tune parameters using the same process. Since the social simplicial complex representation given an f and $\mathbf{P}$ is deterministic, it is particularly well-suited for parameter tuning, as all results and artifacts are reproducible. We demonstrate this as well in Section 5.

5 Evaluation

To validate the benefits of the social simplicial complex, we show the efficacy of the SoTA Toolkit and demonstrate the use of social cohesion and persistence to understand social relationships.

5.1 Validation of the SoTA Toolkit

Here, we show the power of using the social simplicial complex and its measures of social cohesion and social persistence to identify how accurately a function captures the social setting it is applied to. We evaluate two aims: 1) the toolkit accurately identifies the model that better suits each social scene type (Experiment 1); and 2) the toolkit enables model parameter tuning (Experiment 2).

For the first aim, we must show that measures derived from the social simplicial complex accurately reflect the models they represent. In other words, a function designed for a specific dataset type should perform better than a function designed for an entirely different type of setting. We perform validation using two simple functions that separately capture foundational elements of the static social group literature or the dynamic social group literature in combination with two datasets that separately capture static or dynamic group settings. By showing that the SoTA toolkit can accurately identify the model that best matches each dataset, we can demonstrate the efficacy of our method as a tool for designing and evaluating models of social-spatial relationships.

For the second aim, we take the view of a system designer trying to optimize performance on static social group scenes and demonstrate the process of parameter tuning.

5.1.1 Experiment 1 Design.

Datasets: For the static social environment, we use the cocktail party portion of the SALSA dataset [1], which captures scenes of people in a reception setting. For the dynamic social environment, we use a portion of the Groups DIAMOR dataset gathered in an underground passageway between a train station and a mall [13, 55]. We train our functions using 50 scenes from the SALSA dataset and 50 scenes from the Groups DIAMOR dataset in two rounds of development (20 for the first and 30 for the second). We perform the evaluation on a separate test set of 30 scenes from each dataset.

Metrics: The output from the SoTA Toolkit—a ranked list of social groups—can be compared against the ground truth social group labels provided by these datasets. We consider social groups of three or more, which is the maximum of Kendon and Moussaïd's smallest reported group sizes [22, 33]. We evaluate these rankings using the true positive rate (TPR), the number of groups from the ground truth correctly identified in the top k ranked groups proposed divided by the number of total ground truth groups, and the false positive rate (FPR), the number of groups in the top k ranked groups proposed that are not members of the ground truth divided by k . The overall performance is reported by the difference TPR-FPR.

Baseline Models: We introduce two models, one representing static and the other representing dynamic social relationships, that each capture a core principle about social spatial formations from their respective community. To provide the most accurate comparison, we design the models to have minimal differences.

For the static social relationship model, we adopt Kendon's notion of shared, exclusive social space from his work on transactional segments and O-spaces in F-formations [22]. The extensive prior work using Kendon's F-formations [38, 39, 42, 46, 56] informed this model choice. For the dynamic social relationship model, we incorporate Moussaïd et al. [33]'s geometry of walking social groups and parameter values drawn from real-world observations, which align

with formations described by other literature [12, 54]. We formalize these two models below.

Let each person p in a scene be represented by cartesian coordinates (x_p, y_p) , a heading angle represented as unit vector a_p , and a velocity vector v_p . $p = (x_p, y_p, a_p, v_p)$.

We define a point p' that captures the relevant notions of social space for each model. For the static model, let $p' = (x_p + h * a_{px}, y_p + h * a_{py})$ where h is a parameter that defines half of a valid proxemics distance for a pairwise interaction (the other half is given by the other member of the pair). For the dynamic model, $p' = (x_p + q * v_{px}, y_p + q * v_{py})$ where q is a parameter that defines the amount of time to simulate each person's step forward. This incorporates the velocity in the dynamic model but not the static model, effectively handling the differing sets of inputs.

The output of our function captures pairwise social relationships between (p_1, p_2) , defined as

$$f(p_1, p_2) = [x_{p_1}, y_{p_1}, x_{p_2}, y_{p_2}, x'_{p_1}, y'_{p_1}, x'_{p_2}, y'_{p_2}] \quad (1)$$

For both models, we require valid outputs to be generated from a (p_1, p_2) pair that are at most a distance of h apart from each other to account for proxemics [17].

We incorporate Kendon's idea of transactional segments in the static model by filtering for pairs of individuals who are both "in front" of each other (within a 180 degree field of view of the other [24]) and whose pairwise distance decreases as both members are perturbed in the direction they are oriented to reflect "equal and exclusive access" [18]. The dynamic analog is requiring that pairs of people maintain an approximate 90 degree angle between the segment connecting the two and the heading angle of at least one person in the pair. This restriction is drawn directly from Moussaïd et al. [33]'s empirical results, and we run our model with a tolerance of three standard deviations (68.08 to 118.18 degrees) [33] to account for head rotation in addition to heading direction [24]. We also require that velocities of walking pairs not point in opposing directions, i.e., that $v_{p_1} \cdot v_{p_2} \geq 0$.

These models are relatively simple—each capturing foundational elements unique to their respective social scenarios—while minimally differing from each other to demonstrate the social simplicial complex's ability to uncover the underlying topology of a model. Our aim is not to create an optimal model, but instead to verify that each model significantly corresponds to its suited dataset.

5.1.2 Duplicate Group Collapsing. Since our social distance functions model pairwise relationships, certain individuals may be listed in multiple groups, or social groups may be proposed that are subsets of other proposed groups. This overlap is desirable as social relationships are not always one-to-one with membership, such as in the example interpretations of social grouping shown in Figure 4. Having recurring subsets with better social cohesion also increases confidence about the set of people's social entanglement.

However, the ground truth labels from our datasets are not ranked, so we must first remove "duplicate" group proposals. Because we evaluate performance using Setti et al. [44] form of equivalence, which does not guarantee a total ordering for subsets, we provide a collapsing algorithm (included in our open-source code). This algorithm greedily removes groups if they are equivalent to a higher-ranked group based on Hedayati et al. [18], Setti et al.

[44]'s measures of group equivalence. This group collapsing algorithm fits into the framework in a modular fashion, so a designer may choose to alter this process depending on their needs (for example, additionally removing groups that are strict subsets of higher-ranked proposed groups). We note that removing duplicates cannot improve true positives and only limits false positives by not double-counting two incorrect groups that are equivalent.

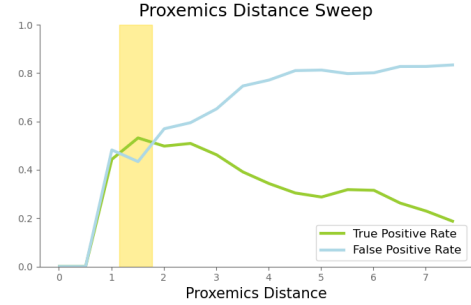


Figure 5: True positive rate and false positive rate as h (representing proxemics distance) varies. The yellow shading shows the optimal parameter window, which corresponds to close social distances for interactions among acquaintances as found in previous literature on proxemics [17].

Condition	SALSA	DIAMOR
Static Model	.059	0
Dynamic Model	-.474	0.344

Table 1: The difference between the true positive rate and false positive rate reported for the static and dynamic models on the static social scene dataset (SALSA) and dynamic social scene dataset (Groups DIAMOR). Orange indicates the superior model.

Results: When running our static model and our dynamic model on the static dataset, we indeed find that the static model outperforms the dynamic model with statistical significance ($p = 1.49e^{-5}$). TPR-FPR is $.539 - .480 = .059$ for the static model and $.231 - .705 = -.474$ for the dynamic model (Table 1).

Furthermore, when running the models on the dynamic dataset, we also find that the dynamic model is better suited with statistical significance ($p = 0.00879$). TPR-FPR is $.067 - .067 = 0$ for the static model and $.500 - .156 = .344$ for the dynamic model (Table 1).

5.2 Parameter Tuning

The default distance parameter h for our static model is 2.1, the value for the close phase of interaction from proxemics literature [15–17]. However, as observed in [33], this parameter is dependent upon the scene density and available space. Below, we show how our framework can also be used to tune this parameter.

We use the static model from above to tune parameter h on 30 scenes of the training data drawn from the SALSA dataset. We find

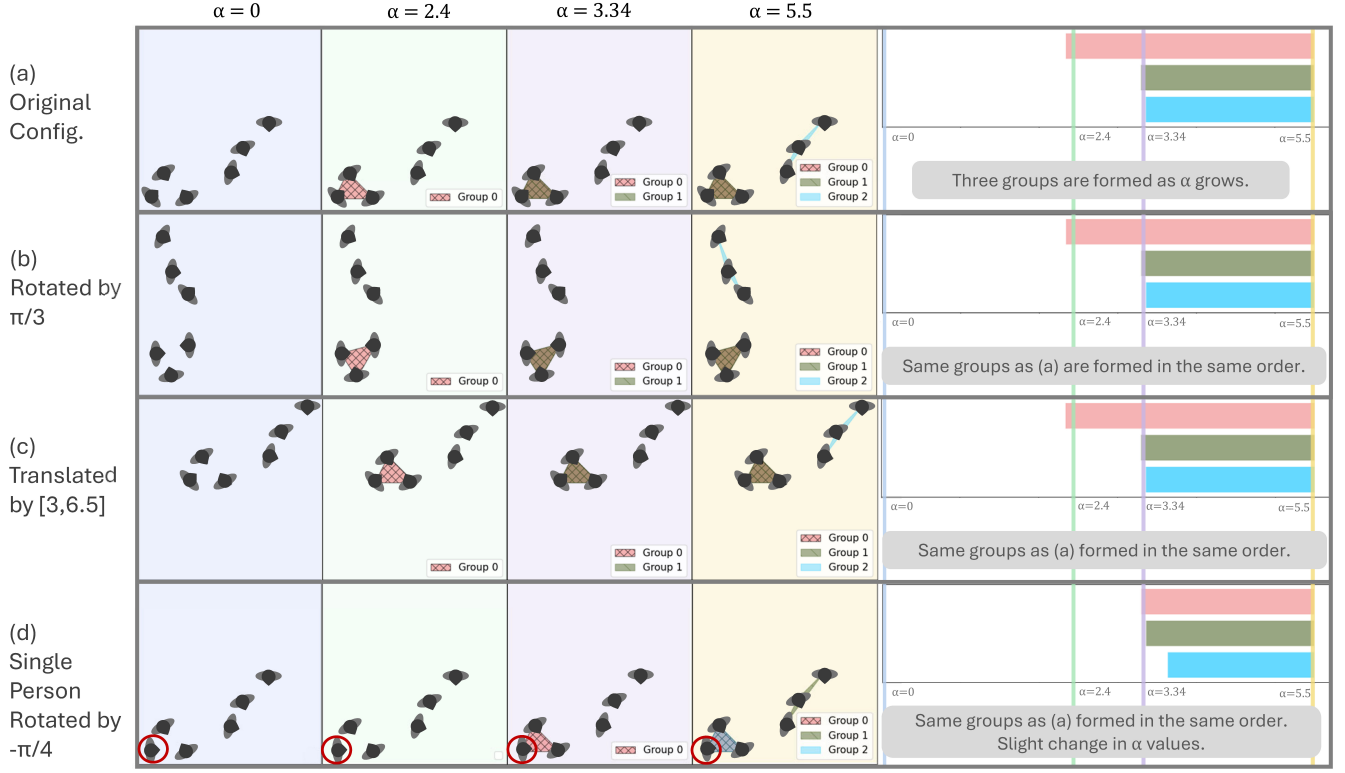


Figure 6: Invariance of proposed groups to geometric perturbations using our method. Left: groups formed in a toy scene as α increases. Right: barcode diagram showing the persistence of groups’ corresponding social simplicial complexes. Rows: (a) original scene, (b) with translation, (c) with rotation, (d) with rotation of one person—circled—slightly away from group center.

that the optimal parameter window, as shown in Figure 5 suggests an h of 1.51, which aligns with expected proxemics distance—the 1.2 to 2.1 meter range for close social distance interactions [17]. If we instead use $h = 1.51$ on the test split, the TPR minus FPR score increases to 0.228 from 0.215.¹

5.3 Topological Invariance and Social Equivalence

Now that we have defined and selected an $f(\mathbf{P})$ that satisfies the conditions described in Section 3, we can revisit the guarantees provided in Section 3.2 and show topological invariance in practice.

To illustrate the ability of our method to remain invariant to geometric changes, we analyze a toy scene with six people: three clearly in a social group and the other three potentially in a loose social interaction Figure 6. We visualize the three most socially-cohesive simplicial complexes with a social persistence value above a “noise” threshold ($\eta = 0.03$) before and after two geometric transformations, translation and rotation, to the people’s positions (a-c). We additionally include one example (d) where a single person is rotated away from the group center, which *should* impact the detected social groups in the scene. This weakening of social relationships is visualized in the associated persistence diagram, where the bars

corresponding to the three-group in the bottom left appear later and are shorter than in the other scenes.

For the translation and rotation perturbations, Figure 6 (b) and (c) show that the two most cohesive simplexes correspond to the same group comprised of three people. The third most cohesive simplex corresponds to the other three people designed to be in a loose interaction. Importantly, these proposed groups remain the same as the original (a) through the translation and rotation transformations. The persistence barcodes on the right of the diagram remain the same as the original through both transformations, reflecting that the structure of the social simplicial complex does not change.

To demonstrate robustness against real-world noise, we investigate two consecutive timesteps of scenes from the SALSA dataset, which reflects both sensor noise and natural shifting of poses as social dynamics in the room evolve (Figure 7). Note that the most highest-ranked (most socially cohesive) two groups and the members identified overall in the top six groups remain the same between both scenes. However, the social persistence of proposed groups changes between the two timesteps (reflected in the length of the barcodes), allowing us to predict future social evolution. In Figure 7, one member of the four-person cluster in the upper center of the scene changes orientations, perhaps to turn away from a conversation. Thus, the corresponding bar (pink) becomes less socially persistent, indicating that the strength of the group’s

¹Note that if we optimize the dynamic model for its own h on the same training scenes, its best parameter of 2.01 still performs significantly worse than the static model ($p = 3.76e - 8$) with a score of -.486.

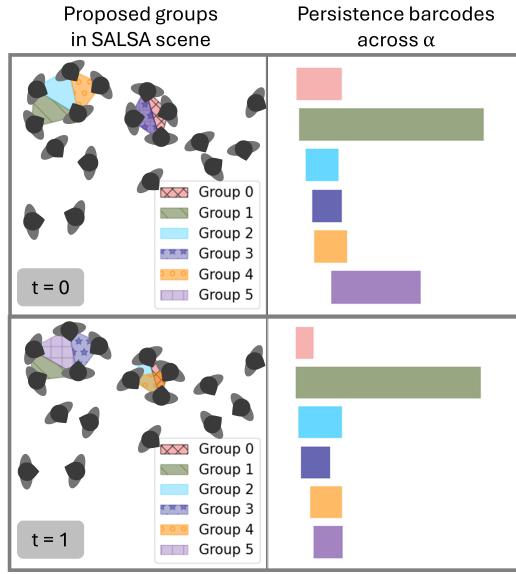


Figure 7: The six most likely proposed groups each of two consecutive scenes drawn from the SALSA dataset (left) with their corresponding barcode diagrams (right). The two most cohesive proposed groups and the members identified over-all in the six most cohesive groups remain the same. The orientation changes of the members of the top center cluster are reflected in social persistence (e.g., in the change to the corresponding pink bar in the barcode diagrams).

social entanglement is not as strong as before, and the four-person group may be in the process of breaking.

6 Takeaways and Future Work

In this work, we introduce a novel mathematical formalism using persistent homology from topological data analysis for understanding social-spatial groups. We show the strengths of the quantifiable metrics derived from this representation for social scene understanding and contribute the open-source SoTA Toolkit.

6.1 Further Applications and Extensions of the Social Simplicial Complex

Notably, the social simplicial structure could be used for simulation and synthetic data generation. Given a set of poses, one could generate its social simplicial complex, then use it to create a variety of new scenes with the same underlying social-topological structure. The simplicial complex could also be modified (e.g., edges removed or added) to generate a scene with dictated social perturbations. This allows for an efficient way to artificially create realistic social scenarios with ground truth data labels.

For robotics, the social simplicial structure can be used to label data to facilitate the training of neural networks for socially-appropriate navigation. Moreover, the numerical measures of social cohesion and persistence can inform real-time algorithms to both adjust to and influence natural changes in social group dynamics.

A simplicial complex that captures social information can be created for any model that captures social information that lies

in a well-defined vector space, not only spatial relationships. For this work, the assumptions and restrictions on the input social relationship function $f(\mathbf{P})$ as defined in Section 3 include that it must reflect spatial relationships, but from a purely mathematical standpoint, our simplicial structure generation process only requires that $f(\mathbf{P})$ is injective and that its output is in vector format. Thus, our formalism can be extended to evaluate models that capture speech volume or gesture, for example.

It is also possible to use dimensions higher than the $0th$ homology group examined in this work. The $0th$ dimension is the most basic use of persistence and only captures a portion of persistent homology’s analytical potential. Using $1D$ persistent homology, we can also analyze loops in social-spatial data to capture social ideas that are not geometrically intuitive, such as how a robot’s actions can change a group’s conversations without even needing to enter it [11]. This method has been applied to other fields such as understanding protein interaction networks [41], where $1D$ loops equate to alternate interaction pathways, or cancer genetics [40], where $2D$ persistent voids select gene removals to alter cancer networks. One limitation is that persistence may be computationally slow, especially with very large datasets; as a result, it would not be suitable for on-board robot use without modification.

6.2 Algebraic Topology as a Powerful Tool for Social-Spatial Techniques

A core contribution of our work is introducing persistent homology to the analysis of human social patterns. Given that people naturally use topology to reason about spatial information in a lower-dimensional representation [25, 26], topological analytical tools are well-suited. Furthermore, algebraic topology includes a rich set of tools outside of persistence that can be used for social-spatial applications, such as homotopy [40] and topological natural language primitives [29, 45] for robot navigation. We are excited about the potential for algebraic topology to enable computationally accessible alignment with human topological notions of space, unlocking novel methods for social-spatial analysis.

6.3 Safe and Responsible Innovation Statement

Societal impact: We build toward more natural social agents and provide tools for developers to explore social information more accessibly. *Ethical considerations:* Any data used with the SoTA toolkit must protect privacy and anonymity. *Responsible Deployment:* Any functions used as input to the SoTA toolkit must not cause harm—social, physical, or otherwise. We do not condone use of this work in a surveillance setting. If extended for generating artificial data in the future, it will be important to address how biases in a priori data may be amplified if it is not rebalanced.

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